UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

NYARUGENGE CAMPUS

COMPUTER AND SOFTWARE ENGINEERING

MODULE: MOBILE APPLICATION SYSTEM AND DEVELOPMENT

On 08th April, 2026

# LAB 6: PUSH NOTIFICATIONS WITH FIREBASE CLOUD MESSAGING (FCM)

# LAB 6: PUSH NOTIFICATIONS WITH FIREBASE CLOUD MESSAGING (FCM)

## Objective.

The objective of this lab is to implement push notifications in a Flutter application using Firebase cloud Messaging (FCM) and Flutter local notifications, and to test notification delivery in foreground, background and terminated states.

## Observations

> Firebase cloud Messaging (FCM) requires a real device; ~~such as~~ emulators do not fully support push notifications.

> On Android 13 or higher, Push Notifications permission must be requested at runtime for notifications to work.

> Background notifications are automatically handled by the system.

> Foreground notifications require Flutter local notifications, as FCM does not display them by default.

> The FCM token is unique per device and may change if the app is re-installed or app data is cleared.

> The google-services.json file is essential; without it, Firebase cannot initialize

> Testing notifications requires an active internet connection, and messages can be sent from Firebase Console.

> Testing on emulators is unreliable; using a real device is recommended.

## Procedure.

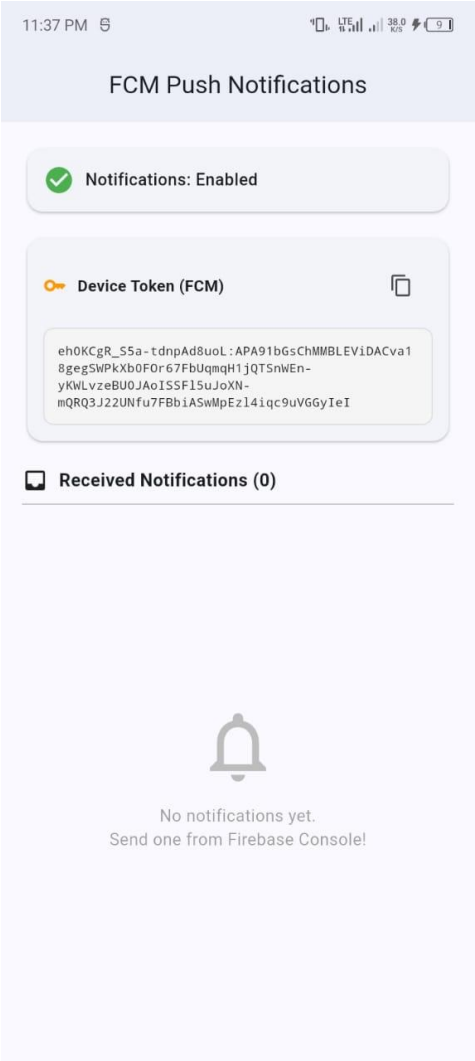
1. Created a Flutter project and added Firebase libraries along with Flutter local notifications.
2. Configured the Firebase project and registered the Android application.
3. Downloaded the google-services.json file and added it to Flutter project in the correct location.
4. Requested notification permissions at runtime.
5. Retrieved FCM device token and displayed it on the screen.

6. Set up listeners to handle notifications in foreground, background, and terminated states.

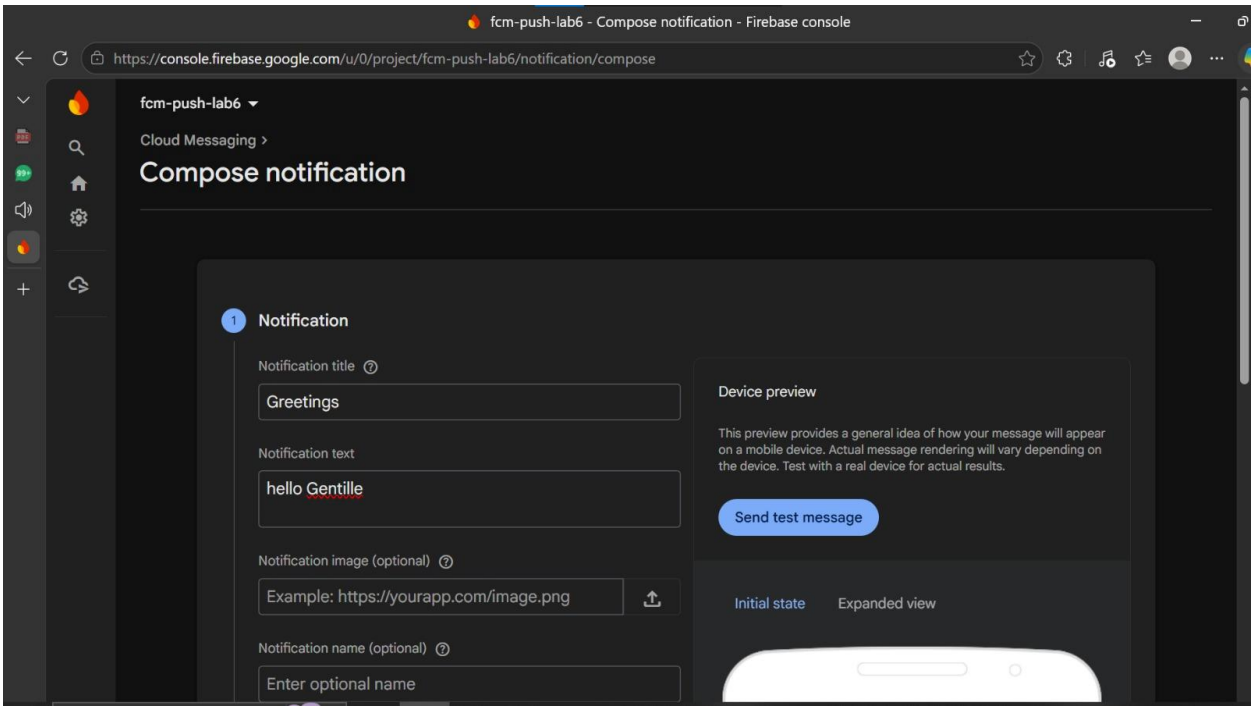
7. Sent test notifications from Firebase console using the device token.

8. Verified that notifications were successfully received in all app states.

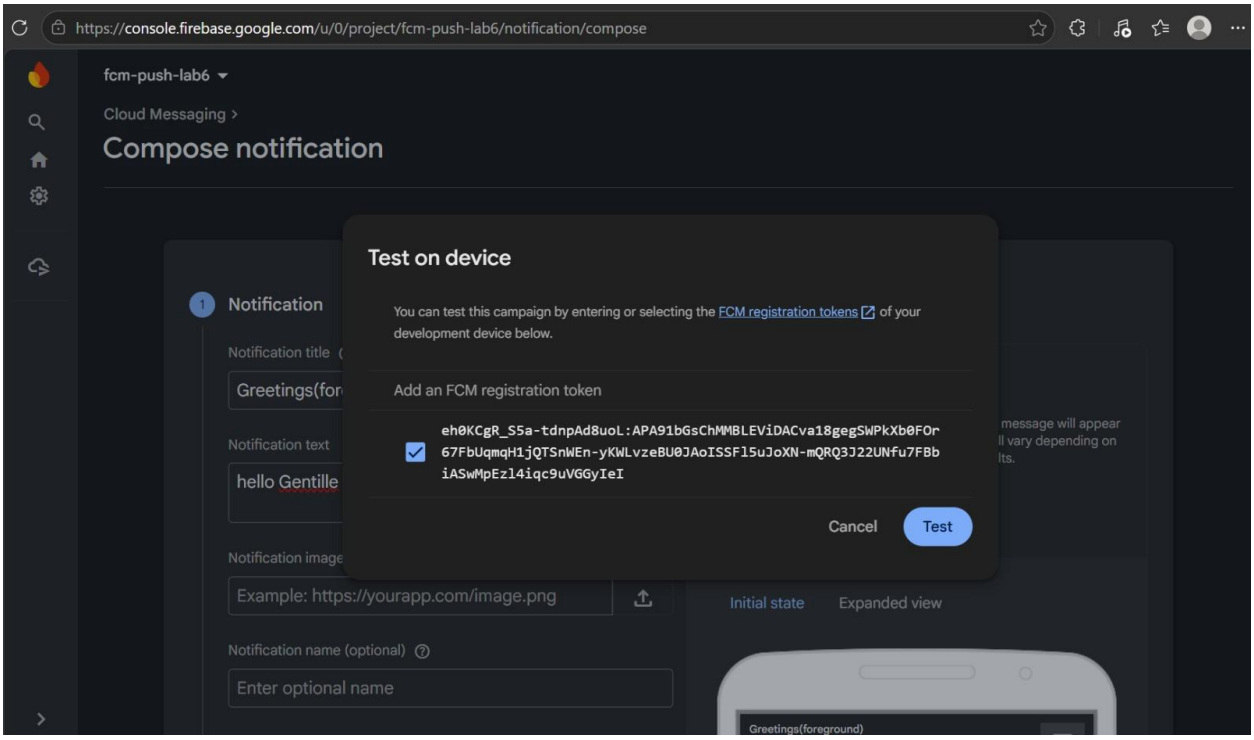
1. Upon opening the app for the very first time on a new device, this beautiful screen appears with the device token ID, and no notification has been received yet.



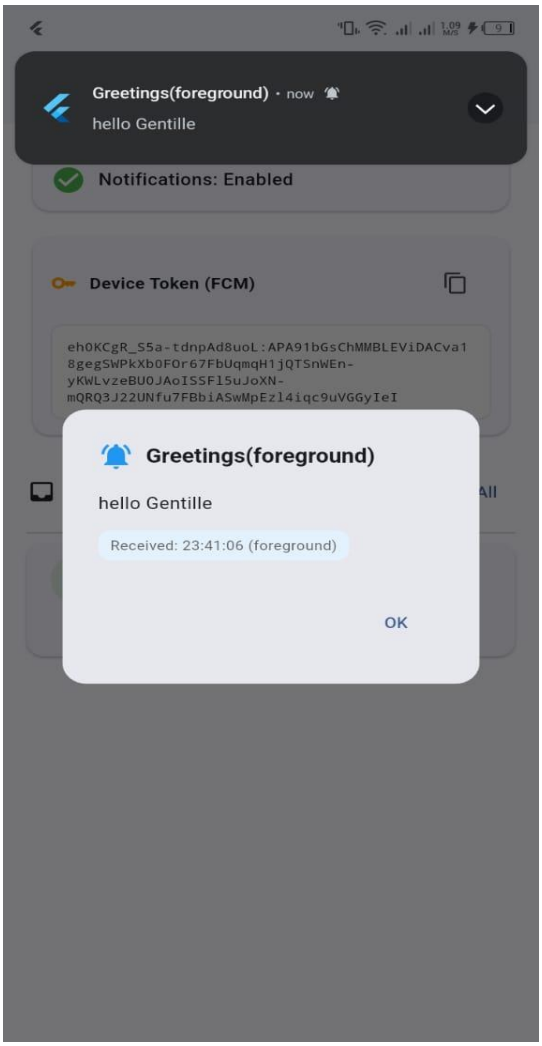

2. **After copying the device token ID, I accessed the Firebase console, went to the Cloud Messaging section, selected Compose Notification, and entered a test message.**

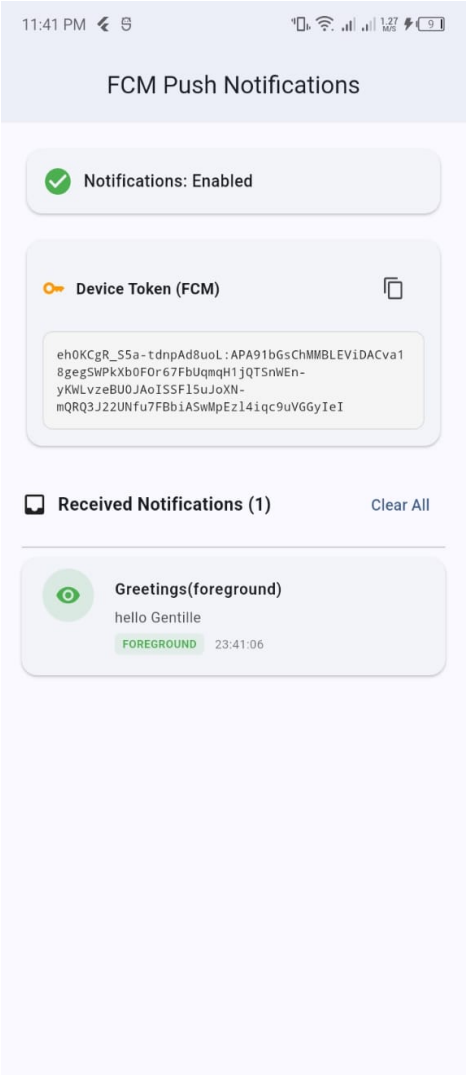



3. After clicking the 'Send text message' button, I entered the device token ID and saved it by checking the box.

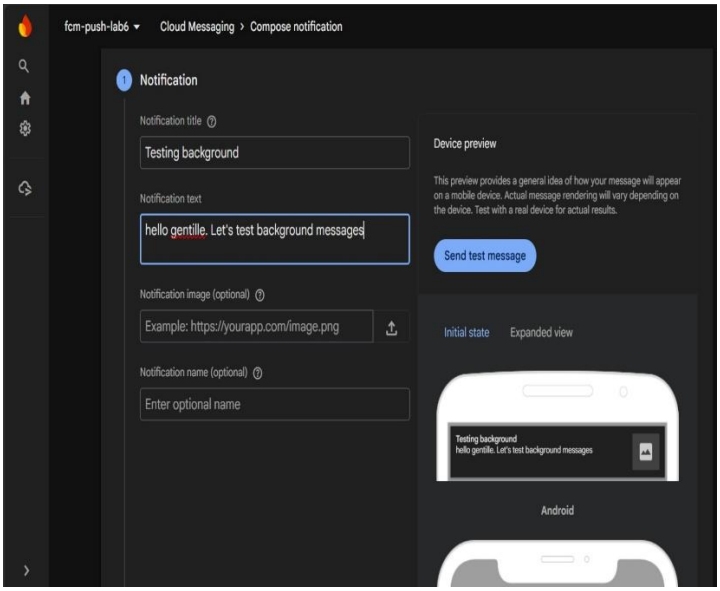


4. **As shown in the screenshots, I received the message on my phone while it was in foreground mode, since the app was open.**

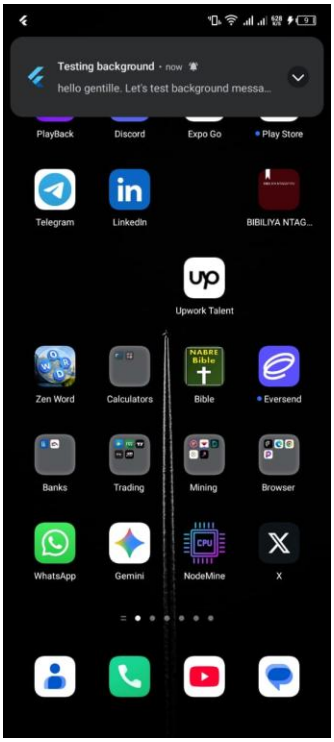


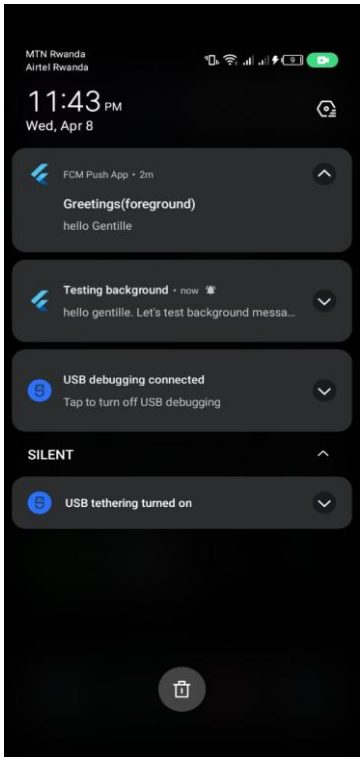



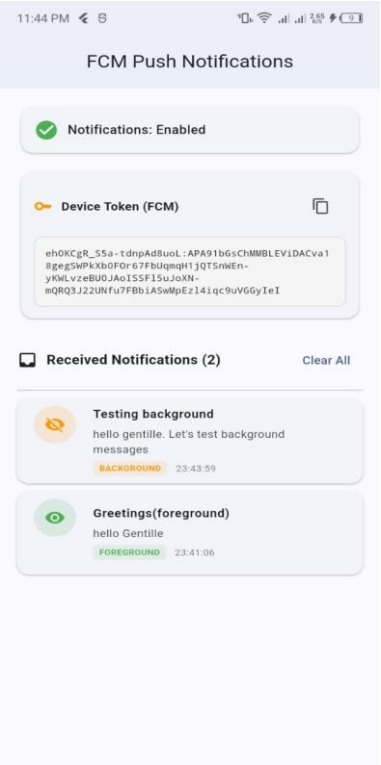

5. Tested Background mode



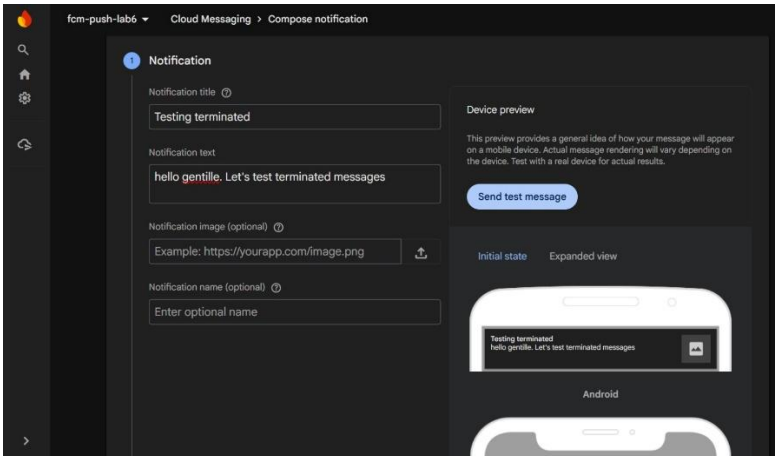

Received it on the device as a background message.

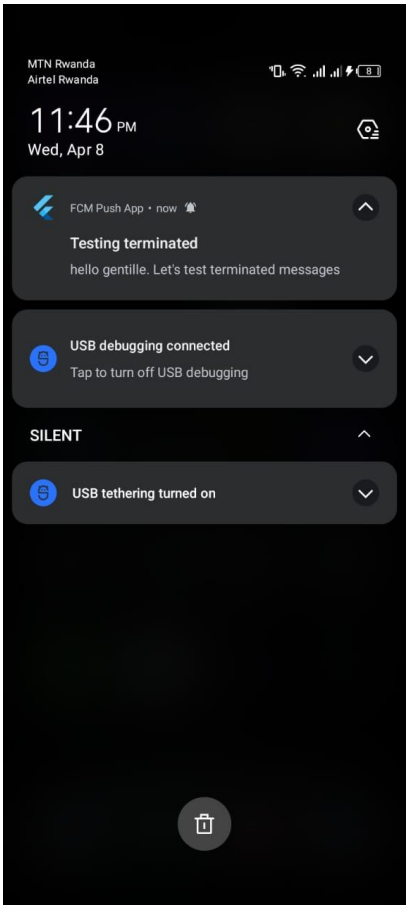


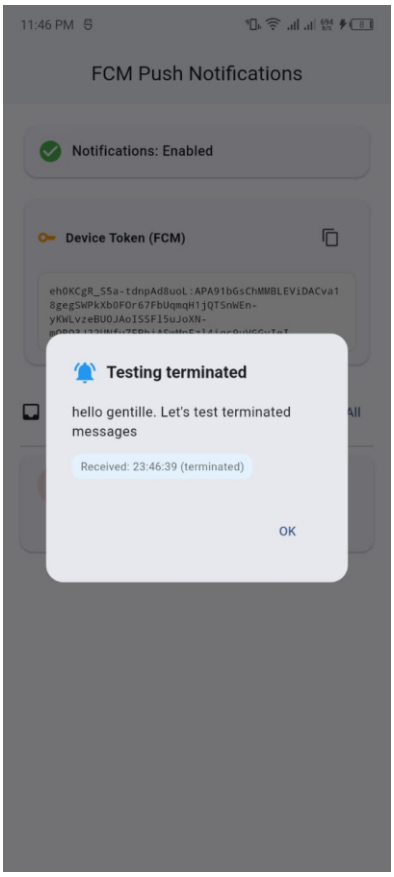



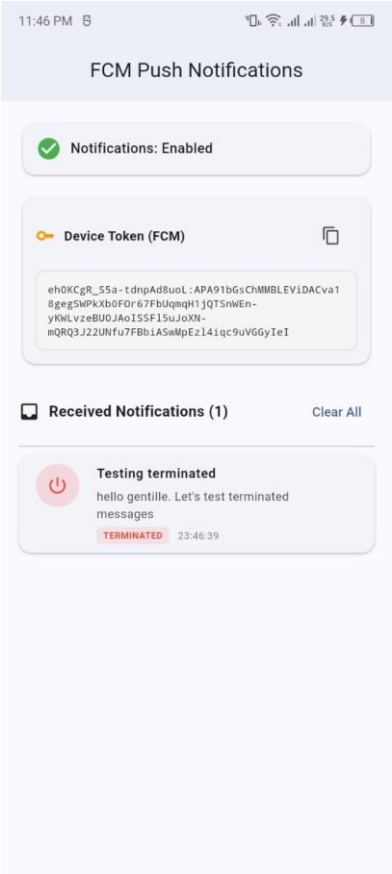



6. To demonstrate the app's behavior, I closed it completely, and it showed a terminated state.




11:46 PM
FCM Push Notifications
Notifications: Enabled
Device Token (FCM)
Testing terminated
hello gentille. Let's test terminated messages
Received: 23:46:39 (terminated)
OK


11:46 PM
FCM Push Notifications
Notifications: Enabled
Device Token (FCM)
eh0KCgR_S5a-tdnpAd8uoL:APA91bGsChMMBLEViDACva18gegSWPkXb0FOr67FbUqmqH1jQTSnWEn-yKWLvzeBU0JAoISSFl5uJoXN-mQRQ3J22UNfu7FBbiASwMpEzl4iqc9uVGGyIeI
Received Notifications (1)
Clear All
Testing terminated
hello gentille. Let's test terminated messages
TERMINATED
23:46:39